Name - Khushi

Btech CSE Core 'A'

Roll no - 2401010115

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Database Management System

Assignment - 02 (Unit - 2)

Section - A →

Q-1 (i) The Entity Relationship Model visually represents how entities relate in a database, offering a clear blueprint of the data's organizational structure.

(ii) In the Relational model, data is organized into tables, with relationships b/w entities defined through foreign key constraints.

(iii) The Object-Oriented model treats data as a collection of objects, each with unique attributes & methods, mirroring real-world entities & interaction.

(iv) Integrity constraints ensure the reliability of data by enforcing rules that must be adhered to, maintaining the accuracy of information.

Section - B →

Q.2 In a healthcare system, discuss how data abstraction can simplify the interaction between healthcare professionals & patient records.

⇒ Data abstraction is the process of hiding

complex database details and showing only relevant information to users.
In a healthcare system, data abstraction helps doctors, nurses & staff easily access patient information without dealing with complex database structures.

For ex:- ↳ Doctors can view patient medical history, diagnosis & prescriptions without seeing how the data is stored in the database.
↳ Nurses can update patient vitals & treatment details.
↳ Receptionists can only access appointment & basic patient details.

Through abstraction levels :-

(i) Physical Level :- Shows how data is stored in the database.
(ii) Logical level :- Shows the structure of the database like tables & relationships.
(iii) View level :- Shows only the data required by specified users.
This makes the system easier to use, secure & efficient for healthcare professionals.

Q-3 Explore the potential benefits & challenges associated with implementing data abstraction in this context.

Benefits :- (i) Security → Sensitive patient information can be restricted to authorized users only.

(ii) Simplicity → Healthcare staff can access required data without understanding database complexity.

(iii) Data Integrity → Maintains accuracy & consistency of patient data.

(iv) Efficiency → Faster access to patient records improves medical-decision making.

Challenges:- (i) Implementation Complexity :- Designing multiple abstraction level can be difficult.

(ii) Performance Overhead :- Additional abstraction layers may slightly affect system performance

(iii) Maintainenace Difficulty → Updating the system requires careful management of different views & structures.

Section-C (Simulation-based)

Q8 Design a database for a university, incorporating entities like students, course & professors. Using the entity-relationship model, illustrate & explain the relationships b/w these entities. Include attributes for each entity to capture relevant information such as student names, course, details & professor information.

⇒ Entities & Attributes :-

(i) Student :-

Attributes :-

- Student_Id (Primary Key)
- Department.
- Name
- Age
- Email.

② Course :-

- Course_ID (PK)
- Course_Name.
- Credits
- Department.

③ Professor :-

- Professor_ID (PK)
- Name
- Email
- Department

Relationship :- <i> Student - Enrolls - Course (M:N)

- One student can enroll in many course.
- One course can have many students.

<ii> Professor - Teaches - Course (1:M)

- One professor can teach multiple courses.
- Each course is taught by one professor.
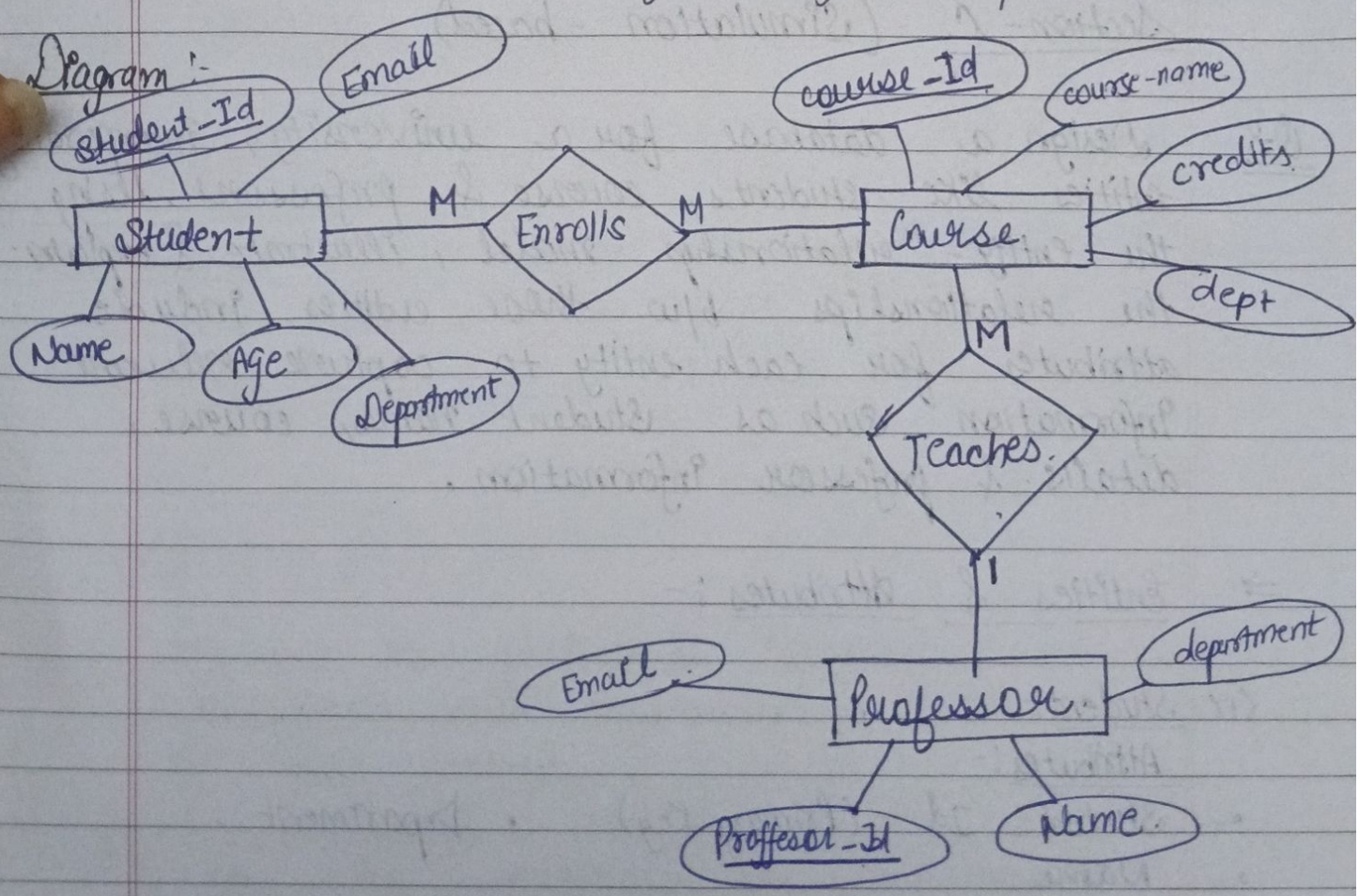
Diagram :-